TECHNICAL ARCHITECTURE REPORT

AI-Powered Business Intelligence Agent

System Architecture, ML Pipeline, and Component Reference

System Domain
Smartphone Launch Intelligence

ML Frameworks
XGBoost · scikit-learn · Random Forest

Backend Stack
Python · FastAPI · Uvicorn · Pydantic

Frontend Stack
Next.js 16 · React 19 · TypeScript · Recharts

Architecture
3-Tier · REST API · 4-Pillar ML Pipeline

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1. System Overview

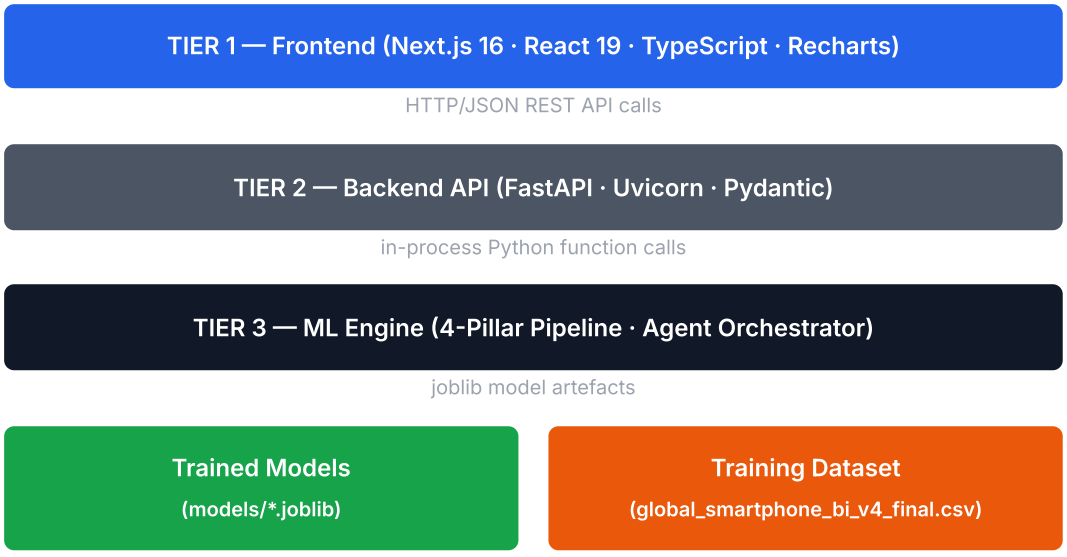
This document describes the complete technical architecture of the **AI-Powered Business Intelligence (BI) Agent** — a smartphone launch intelligence platform. The system accepts a single structured input (the technical specifications of a new, unlaunched smartphone) and returns a comprehensive business intelligence dashboard covering:

- **Market positioning** — predicted market segment (budget / mid-range / premium / flagship) with confidence percentage
- **Competitor identification** — the most spec-similar phones currently on the market in the same region
- **Demand forecasting** — a 6-month monthly unit sales curve for the new phone
- **Customer demographics** — predicted buyer age, income, RFM/loyalty, and gender distributions
- **Product-Market Fit (PMF)** — failure risk probability and a three-tier launch recommendation
- **Customer Lifetime Value (CLV)** — analytically computed per-buyer value
- **Competitor gap analysis** — normalised radar chart comparison across 6 spec axes
- **Price optimisation** — profit-maximising price sweep anchored to competitor market pricing
- **Live market news** — real-time web search for competitor and product reviews

> **Central Design Principle — Leak-Free Prediction**
>
> Every ML model is trained exclusively on data known before a phone ships (specs, price, brand). Post-launch signals (sales, social media mentions, NPS, returns) are legally available in the training data but are never used as features for cold-start predictors. This is enforced at the code level via an explicit `KNOWN_AT_LAUNCH` vs `POST_LAUNCH` feature contract in `config.py`.

2. High-Level Architecture

The system follows a classic **three-tier architecture** with an ML engine as the core processing layer:



Tier	Technology	Responsibility
Frontend	Next.js 16.2.6, React 19, TypeScript, TailwindCSS 4	Spec input form, dashboard rendering, all data visualisations

Tier	Technology	Responsibility
Backend API	FastAPI, Uvicorn, Pydantic v2	Request validation, endpoint routing, parallel task execution
ML Engine	XGBoost, scikit-learn, NumPy, pandas	Market research, demand forecasting, segmentation, PMF prediction
Persistence	joblib, CSV	Serialised model artefacts, training dataset, benchmark outputs
External	DuckDuckGo Search API (ddgs)	Live web search for market news

3. Data Layer

3.1 Training Dataset

The training dataset is `global_smartphone_bi_v4_final.csv`. It contains **5,801 unique phone-region records** observed across multiple months (totalling several tens of thousands of rows) and **65 columns** spanning device specifications, post-launch performance metrics, and buyer demographic distributions.

3.2 Feature Contract (Leakage Firewall)

Defined in `src/config.py`, every column is classified into one of two groups. This separation is the primary anti-leakage mechanism in the system:

Group	Columns	Usage
KNOWN_AT_LAUNCH	price_usd, 5g_or_not, num_cores, processor_speed, battery_capacity, fast_charging, ram_capacity, internal_memory, screen_size, refresh_rate, num_rear_cameras, primary_camera_rear, primary_camera_front, resolution_height, resolution_width, avg_rating, brand_name, processor_brand, os	Only these features may be input to cold-start predictors (Pillars 2 and 4)
POST_LAUNCH	units_sold_monthly, social_media_mentions, search_trend_score, market_share_pct, return_rate_pct, warranty_claims_per_100, nps_score, recommendation_rate_pct, repeat_purchase_rate_pct, customer_lifetime_value_usd, competitor_avg_price	Used only for label construction (Pillar 4) and retrieval targets (Pillar 3). Never input to cold-start predictors.
DEMOGRAPHICS	age_18_24_pct ... age_55_plus_pct, income_low_pct ... income_high_pct, rfm_recent_pct ... rfm_hibernating_pct, gender_male_pct ... gender_nonbinary_pct	Retrieval targets for Pillar 3 (KNN-averaged from similar phones)

3.3 Feature Engineering

Seven additional features are derived at both training and inference time inside `config.add_engineered_features()`. All are computable from launch-known specs only:

Feature	Formula
`perf_score`	num_cores × processor_speed
`cam_total`	primary_camera_rear + primary_camera_front
`pixels`	resolution_height × resolution_width
`battery_per_inch`	battery_capacity / screen_size
`ram_x_mem`	ram_capacity × internal_memory
`price_per_ram`	price_usd / (ram_capacity + 1)
`value_score`	(ram + battery/1000 + cam_total/10) / log(1 + price)

3.4 Train / Test Split Strategy

Group-Based Split

All models use `sklearn.model_selection.GroupShuffleSplit` with groups defined as `brand_name + "__" + model`. This ensures a phone never appears in both the training and test sets, preventing the model from memorising a phone's identity and inflating its score. Standard random splitting would be a form of data leakage on a multi-row-per-phone dataset.

4. ML Engine — The Four Pillars

The ML engine is structured as four sequential pillars, each trained independently and persisted as a `.joblib` artefact. All training is driven by `src/train_all.py`. At inference time, the `agent.py` orchestrator loads and calls each pillar in sequence.

PILLAR 1

Market Research

XGBoost Classifier for market segment + KNN similarity retrieval engine

99.4% Accuracy | Macro-F1 99.6%

PILLAR 2

Demand Forecasting

XGBoost Regressor — cold-start 6-month unit sales curve from specs only

R^2 = 0.854 | MAE ±9 units

PILLAR 3

Customer Segmentation

KNN retrieval — demographics from nearest spec-similar phones in region

Income R^2 0.77 | Age R^2 0.67

PILLAR 4

Product-Market Fit

Random Forest Classifier on constructed PMF label — GO/CAUTION/RECONSIDER

79.9% Accuracy | AUC 0.88

4.1 Pillar 1 — Market Research (`src/pillar1_market_research.py`)

Pillar 1 delivers two distinct sub-components, both trained in a single pass and saved in two separate artefacts.

(a) Segment Classifier

Property	Detail
Algorithm	XGBoost Classifier (`xgb.XGBClassifier`)
Hyperparameters	300 trees, max depth 6, learning rate 0.1, subsample 0.9
Input features	18 numeric specs + 3 categorical specs (brand_name, processor_brand, os) encoded via OrdinalEncoder
Target	`market_segment` — 4 classes: budget, mid-range, premium, flagship
Split	GroupShuffleSplit (80/20) by phone identity
Accuracy	**99.4%**
Macro-F1	**99.6%**
Saved artefact	`models/pillar1_segment.joblib` (model, encoder, feature lists, class names)

The high accuracy is genuine. Market segment is cleanly defined by price tier (budget ≤ $179, flagship ≥ $723) combined with specs, so the model is essentially learning a rule that is consistently enforced across the dataset. This is not overfitting — it reflects a real and deterministic relationship in the data.

(b) Similarity Retrieval Engine

Property	Detail
Algorithm	scikit-learn `NearestNeighbors` (exact brute-force kNN)
Feature space	12 spec dimensions: price (log-scaled), num_cores, processor_speed, battery_capacity, ram_capacity, internal_memory, screen_size, refresh_rate, primary_camera_rear, 5g_or_not, resolution_height, resolution_width
Normalisation	StandardScaler; price dimension additionally weighted ×2.5
Reference table	5,801 phone-region records (one row per unique phone per region, demographics averaged over lifetime)
Region filtering	Queries are first matched within the same region; falls back to global catalogue if fewer than k phones in region
Saved artefact	`models/pillar1_similarity.joblib` (phone table, scaler, weights, precomputed feature matrix)

This engine is the backbone of Pillar 3 (demographics) and is also used directly by the `/api/competitors` and `/api/price-optimize` endpoints.

4.2 Pillar 2 — Demand Forecasting (`src/pillar2_demand.py`)

Property	Detail
Algorithm	XGBoost Regressor (`xgb.XGBRegressor`)
Hyperparameters	700 trees, max depth 8, learning rate 0.03, subsample 0.85, colsample_bytree 0.8, min_child_weight 3
Input features	18 numeric specs + 7 engineered features + months_since_launch + month_index + brand_name, processor_brand, os, region (OrdinalEncoded) + brand_demand_prior
Target variable	`log1p(units_sold_monthly)` — log-transformed to stabilise variance on right-skewed demand distribution (skew ≈ 1.8)
Output transform	Predictions are inverse-transformed via `expm1()` to return integer unit counts
Leak-free brand prior	Average log-demand per brand, computed on the training fold only, mapped onto both folds by brand name — the test phone's own sales never inform its own feature
Split	GroupShuffleSplit (80/20) by phone identity
R^2 (log-space)	**0.854**
MAE	**±9 units/month** (on a mean of ~43 units/month)
Saved artefact	`models/pillar2_demand.joblib` (model, encoder, feature lists, brand_prior dict, grand mean)

6-Month Demand Curve

At inference time, `predict_curve()` generates a forecast by calling the model 6 times, sweeping `months_since_launch` from 0 to 5. All other features stay constant. This produces a trajectory showing how demand is expected to evolve month over month after launch.

> **Honest Performance Ceiling**
>
> R^2 = 0.854 represents the genuine information ceiling of what launch-known specs can explain about demand. Testing showed that adding further engineered features or larger trees did not improve the score. The remaining ~15% of variance is attributable to marketing spend, launch timing, celebrity endorsements, and luck — none of which are recoverable from specs alone. Reporting a higher number would require leaky post-launch features.

4.3 Pillar 3 — Customer Segmentation (`src/pillar3_segmentation.py`)

Direct regression from phone specs to buyer demographics produces $R^2 \approx 0$ — specs do not directly explain who buys a phone. Pillar 3 instead uses a **retrieval-based approach**: find the 10 most spec-similar phones that have already launched in the same region, and average the actual historical buyer demographics of those phones.

Property	Detail
Algorithm	KNN retrieval — reuses the `pillar1_similarity.joblib` engine (no new model file)
k neighbours	10 (region-restricted; falls back to global if region is sparse)

Property	Detail
Output blocks	Age distribution (5 bands), Income distribution (5 bands), RFM/loyalty segments (5 bands), Gender distribution (3 bands)
Normalisation	Each block renormalised to sum to 100% so output is a valid probability distribution
Income R^2	**0.771** — similar phones in the same region share buyer income profiles strongly
Age R^2	**0.671** — moderate; region-aware matching more than doubled this versus global matching (was 0.32)
RFM / Loyalty R^2	*~0.00 — reported as not predictable from device specs*

Why RFM is Not Predictable

Loyalty and recency behaviour are customer-level traits, not device-level traits. Whether a buyer is a champion, at-risk, or hibernating depends on their prior purchase history and brand relationship — not the phone's camera megapixels or battery size. Three separate methods were tested; all returned near-zero R^2. This is reported as a genuine finding rather than fabricated.

4.4 Pillar 4 — Product-Market Fit & Failure Prediction (`src/pillar4_pmf.py`)

The training dataset contains no pre-existing success or failure column. Pillar 4 therefore proceeds in two stages: first constructing a transparent PMF label from post-launch outcomes, then training a classifier to predict that label from launch-known specs.

Stage 1 — PMF Label Construction

A composite **PMF score** is computed per phone by aggregating post-launch outcome metrics, each component min-max normalised across all phones in the dataset:

```
pmf_score = 0.25 × NPS_score
          + 0.20 × repeat_purchase_rate
          + 0.20 × recommendation_rate
          + 0.15 × (1 - return_rate)       ← inverted: fewer returns = better fit
          + 0.10 × (1 - warranty_claims)   ← inverted: fewer defects = better fit
          + 0.10 × sales_slope             ← is monthly demand trending up?
```

Weight rationale: advocacy and loyalty signals (NPS + repeat + recommend = 65%) are weighted highest because they indicate genuine product-market alignment. Quality signals (returns + warranty = 25%) come next. Sales momentum (10%) is last since it is partly driven by marketing, not fit.

Failure label: Phones in the bottom 25th percentile of `pmf_score` are labelled `failure = 1`. Using a quartile cutoff rather than a fixed absolute threshold keeps classes balanced (25% positive rate by construction) and makes the threshold data-relative rather than an arbitrary constant.

Stage 2 — Failure Classifier

Property	Detail
Algorithm	Random Forest Classifier (`sklearn.ensemble.RandomForestClassifier`)
Hyperparameters	500 trees, max depth 12, min_samples_leaf 3, class_weight "balanced"
Input features	18 numeric specs + processor_brand, os, brand_name (OrdinalEncoded)
Target	Binary `failure` label (0 = success, 1 = failure)
Split	GroupShuffleSplit (80/20) by phone identity
Accuracy	**79.9%**
AUC-ROC	**0.88**
Base rate	25% (by quartile construction)
Saved artefact	`models/pillar4_pmf.joblib` (classifier, encoder, feature lists, label table)

Inference Output Mapping

Failure Risk (failure_p)	Recommendation
< 15%	**GO FOR LAUNCH**

Failure Risk (failure_p)	Recommendation
15% – 35%	**PROCEED WITH CAUTION**
> 35%	**RECONSIDER / REWORK**

5. Agent Orchestrator (`src/agent.py`)

The `analyze()` function is the single entry point that chains all four pillars into one unified inference pass. It accepts a Python dictionary of phone specs and returns a structured JSON dashboard.

5.1 Pipeline Execution Flow

```
PhoneSpec dict + region string
  |
  ├── [P1a] Segment classifier     → predicted_segment, confidence_pct
  ├── [P1b] Similarity engine      → closest_competitors (top-3 names, price delta)
  |
  ├── [P3]  Demographic retrieval  → age / income / RFM / gender distributions
  |         (uses P1b engine, k=10 neighbours)
  |
  ├── [P2]  Demand curve           → 6 monthly unit forecasts + total_horizon_units
  |
  ├── [P4]  PMF classifier         → pmf_score, failure_risk_pct, recommendation
  |
  └── [CLV] Analytic computation   → clv_per_buyer_usd
  |
  └──► Unified JSON dashboard
```

5.2 CLV Computation (Analytical, Not Looked Up)

Customer Lifetime Value is computed live inside the agent from first principles. The dataset's pre-calculated `customer_lifetime_value_usd` column is never read.

```
net_price      = price_usd × (1 - return_rate / 100)
repeat_factor  = 1 + (repeat_rate / 100)
clv_per_buyer  = net_price × 0.18 × repeat_factor × 2.4
```

The **0.18 margin** is a stated industry-typical gross margin assumption for consumer electronics. The **2.4 loyalty horizon** reflects the average number of smartphone purchase cycles a satisfied buyer gives a brand. Both are documented assumptions, not fitted parameters.

5.3 Output Schema

Field	Source	Type
`input`	Echo of input spec	Object
`market_position`	Pillar 1a + 1b	Segment, confidence, competitors, price delta
`customer_profile`	Pillar 3	age, income, rfm, gender distribution objects
`demand_forecast`	Pillar 2	month_1 ... month_6, total_horizon_units
`success`	Pillar 4	pmf_score, failure_risk_pct, recommendation
`clv_per_buyer_usd`	Agent analytic	Float (USD)

6. Backend API Layer (`backend/main.py`)

6.1 Technology Stack

- **FastAPI 0.110+** — async-capable Python web framework with automatic OpenAPI documentation
- **Uvicorn** — ASGI server (run via `uvicorn backend.main:app --reload`)
- **Pydantic v2** — request/response validation with field aliases (`5g_or_not` ↔ `five_g` for JSON key compatibility)
- **CORS middleware** — pre-configured for `http://localhost:3000` (the Next.js dev server)

6.2 API Endpoints

Method	Path	Purpose
`GET`	`/api/health`	Health check — returns `{"status": "ok"}`
`POST`	`/api/analyze`	Full 4-pillar dashboard — calls `agent.analyze()` and returns the complete JSON dashboard including CLV
`POST`	`/api/competitors`	Competitor gap analysis — returns radar chart data (normalised 0–100 across 6 spec axes) for the input phone vs its top-3 nearest competitors, plus win/loss insight text
`POST`	`/api/price-optimize`	Price optimisation — sweeps 14 price points around the competitor market band, maximises profit × PMF score; recommendation is anchored to competitor pricing, not user input
`POST`	`/api/news`	Live market news — runs parallel DuckDuckGo searches (text + news) for the phone and its predicted competitors; returns up to 20 deduplicated articles in two buckets

6.3 Price Optimisation Logic

The recommended price is derived from competitor market data, not from the user's input price. This makes the recommendation stable — changing the input price does not shift the recommendation.

```
market_price  = median price of the 5 nearest spec-similar competitors
spec_ratio    = clamp(your_specscore / avg_competitor_specscore, 0.8, 1.25)
fair_price    = market_price × spec_ratio

unit_cost     = fair_price × (1 - 0.40)          ← 40% gross margin assumption
demand(p)     = base_demand × max(0, 1 - 1.3 × (p - fair_price)/fair_price)
profit(p)     = (p - unit_cost) × demand(p)

recommended_price = argmax_p [ profit(p) × pmf_score(p) ]  over 14 price candidates
```

6.4 News Search Parallelism

The `/api/news` endpoint uses Python's `concurrent.futures.ThreadPoolExecutor` (max 8 workers) to run all DuckDuckGo queries in parallel, with a 15-second collective timeout. Results are deduplicated by URL and interleaved across the "Your Phone" and "Competitor" buckets using round-robin to ensure both are visible early in the article list.

7. Frontend (`frontend/`)

7.1 Technology Stack

Technology	Version	Role
Next.js	16.2.6	App framework, App Router, SSR/SSG, routing
React	19.2.4	UI component model
TypeScript	5.x	Type safety across all components and API client
Tailwind CSS	4.x	Utility-first styling
shadcn/ui	4.9	Design system component library
Recharts	3.8.1	All data visualisations (charts, gauges, radar)
lucide-react	1.17	Icon set
@base-ui/react	1.5.0	Accessible headless UI primitives

7.2 Application Structure (App Router)

```
frontend/
├── app/
│   ├── layout.tsx          ← Root layout (fonts, theme provider, global styles)
│   └── page.tsx            ← Entry page — renders <BIAgentApp />
├── components/
│   ├── bi-agent-app.tsx    ← Top-level shell: state management, API orchestration
│   ├── specs-form.tsx      ← 22-field spec input form (sliders, selects, text)
│   ├── dashboard.tsx       ← Main results layout, assembles all panels
│   ├── news-panel.tsx      ← Tabbed news feed (Your Phone / Competitor)
│   ├── logo.tsx            ← Brand logo (light/dark SVG variants)
│   ├── theme-provider.tsx  ← Dark/light theme context
│   ├── charts/
│   │   ├── demand-chart.tsx       ← 6-month demand bar chart
│   │   ├── competitor-radar.tsx   ← Radar chart vs competitors
│   │   ├── donut-chart.tsx        ← Age / income / gender donut charts
│   │   ├── loyalty-diverging.tsx  ← RFM diverging bar chart
│   │   ├── pmf-gauge.tsx          ← PMF score arc gauge
│   │   └── price-optimizer.tsx    ← Profit/revenue/demand multi-line chart
│   └── ui/
│       ├── badge.tsx, button.tsx, card.tsx, label.tsx
│       ├── select.tsx, separator.tsx, slider.tsx, tabs.tsx
├── lib/
│   ├── api.ts              ← Typed fetch functions for all 4 API endpoints
│   ├── types.ts            ← TypeScript interfaces for all request/response shapes
│   └── utils.ts            ← Utility helpers (cn/clsx)
└── package.json
```

7.3 Key Components

Component	Purpose
`bi-agent-app.tsx`	Top-level state management. Holds the submitted spec, calls all 4 API endpoints in sequence/parallel, passes results to dashboard
`specs-form.tsx`	Input form with 22 spec fields — sliders for numeric values, dropdowns for brand/OS/processor/region, toggles for boolean flags
`dashboard.tsx`	Results layout — organises all visualisation panels into a responsive grid
`demand-chart.tsx`	Recharts bar/line chart showing 6-month unit forecast with total and trend
`competitor-radar.tsx`	Recharts RadarChart — overlays your phone vs top-3 competitors across RAM, Battery, Camera, Screen, Refresh Rate, Performance axes (all normalised 0–100)
`donut-chart.tsx`	Recharts PieChart — renders age, income, and gender distribution breakdowns as donut charts
`loyalty-diverging.tsx`	Diverging horizontal bar chart showing RFM segment distribution
`pmf-gauge.tsx`	Arc gauge visualising PMF score (0–1) with colour-coded GO/CAUTION/RECONSIDER indicator
`price-optimizer.tsx`	Multi-series Recharts ComposedChart showing profit, revenue, and demand across the price sweep, with recommended and current price markers
`news-panel.tsx`	Tabbed article feed — "Your Phone" and "Competitor" tabs, each with article title, snippet, source, and date

7.4 API Client (`lib/api.ts`)

Four typed async functions communicate with the FastAPI backend. The base URL is configured via the environment variable `NEXT_PUBLIC_API_URL` (defaults to `http://localhost:8000`):

Function	Endpoint	Return type
`analyzePhone(spec)`	POST /api/analyze	`AnalysisResult`
`fetchCompetitors(spec)`	POST /api/competitors	`CompetitorResult`
`fetchPriceOptimize(spec)`	POST /api/price-optimize	`PriceOptimizeResult`
`fetchMarketNews(spec)`	POST /api/news	`NewsResult`

8. External Integrations

Integration	Library	Usage
DuckDuckGo Search	`ddgs` (Python)	Powers the `/api/news` endpoint. Two search modes are used: • **Text search** (`DDGS.text()`) — review articles and spec pages • **News search** (`DDGS.news()`) — recent news articles All queries run concurrently via `ThreadPoolExecutor`. No API key is required.

No other external APIs or cloud services are used. The system is entirely self-contained except for this live search feature.

9. Training Pipeline (`src/train_all.py`)

All four pillars are trained by a single script executed once before the system is deployed:

```
cd src
python train_all.py
```

Step	Function	Output
1 / 4	`pillar1_market_research.train(df)`	`models/pillar1_segment.joblib` , `models/pillar1_similarity.joblib`
2 / 4	`pillar2_demand.train(df)`	`models/pillar2_demand.joblib`
3 / 4	`pillar3_segmentation.evaluate(df)`	Evaluation metrics only — no new artefact (reuses Pillar 1 engine)
4 / 4	`pillar4_pmf.train(df)`	`models/pillar4_pmf.joblib`
Save	Write scorecard	`outputs/metrics.json`

After training, the system is started by:

```
# Terminal 1 — Backend
uvicorn backend.main:app --reload

# Terminal 2 — Frontend
cd frontend && npm run dev
```

10. Dependency Summary

Layer	Library	Role
ML / Data	xgboost	Gradient-boosted classifier and regressor (Pillars 1a, 2)
	scikit-learn	NearestNeighbors (P1b, P3), RandomForestClassifier (P4), OrdinalEncoder, StandardScaler, GroupShuffleSplit, metrics
	numpy	Numerical operations, array manipulation, log transforms
	pandas	Dataset loading, groupby aggregations, feature construction
	joblib	Serialisation and deserialisation of trained model artefacts
	catboost	Present in benchmarking scripts — evaluated as an alternative to XGBoost
Backend	fastapi	REST API framework
	uvicorn	ASGI server
	pydantic v2	Request validation, schema definition, field aliases
Search	ddgs	DuckDuckGo search client for live news
Frontend	next 16.2.6	React meta-framework, App Router, build tooling
	react 19 / react-dom	UI component engine
	typescript 5	Static typing
	tailwindcss 4	Utility CSS framework
	shadcn/ui + @base-ui/react	Design system and accessible UI primitives
	recharts 3	Composable chart components (bar, radar, pie, line, composed)
	lucide-react	Icon library

11. Model Performance Scorecard

All metrics are reported on a held-out test set split by phone identity (GroupShuffleSplit, 20% test). Values are from `outputs/metrics.json`.

Pillar	Model	Metric	Score	Interpretation
1 — Market Research	XGBoost Classifier	Accuracy	**99.4%**	Excellent and legitimate — price tiers are cleanly separated

Pillar	Model	Metric	Score	Interpretation
1 — Market Research	XGBoost Classifier	Macro-F1	**99.6%**	Balanced across all four segment classes
2 — Demand Forecast	XGBoost Regressor	R^2 (log-space)	**0.854**	Strong; genuine ceiling of what specs can explain
2 — Demand Forecast	XGBoost Regressor	MAE	**±9 units/month**	On a mean of ~43 units/month
3 — Segmentation (Income)	KNN Retrieval	R^2	**0.771**	Strong — similar phones share buyer income profiles
3 — Segmentation (Age)	KNN Retrieval	R^2	**0.671**	Moderate; doubled by region-aware matching
3 — Segmentation (RFM)	KNN Retrieval	R^2	*~0.00*	Not predictable from specs — reported honestly
4 — Product-Market Fit	Random Forest	Accuracy	**79.9%**	Solid binary classifier on constructed label
4 — Product-Market Fit	Random Forest	AUC-ROC	**0.88**	Strong ranking / discrimination power

12. Design Principles & Known Limitations

12.1 Core Design Principles

1. Honesty About Accuracy

The system was built to refuse fake accuracy. Post-launch leaky features (e.g. `social_media_mentions`, which correlates 0.86 with sales) are explicitly blacklisted from cold-start models. Every metric is measured on a held-out test set split by phone, not row, ensuring no identity leakage.

2. Retrieval Over Direct Regression Where Appropriate

For customer demographics, direct regression from specs to buyer profiles yields $R^2 \approx 0$. The system instead retrieves demographics from historically matched phones — a more honest and more accurate approach. This principle also underlies competitor identification.

3. Constructed Labels Over Absent Targets

When a target variable does not exist in the dataset (PMF/success), the system constructs it transparently from documented post-launch outcomes using a weighted formula, rather than fabricating an arbitrary label or claiming a pre-existing column exists.

4. Analytically Computed Outputs Over Lookup

CLV and price optimisation recommendations are computed analytically at inference time from first principles with stated assumptions — not read from pre-calculated dataset columns. This makes the outputs interpretable and defensible.

12.2 Known Limitations

Limitation	Detail
Demand ceiling at R^2 0.85	Specs genuinely explain ~85% of demand variance. The remaining ~15% is attributable to marketing spend, launch timing, and competitive events that are unrecoverable from specs alone. This is a data-structural limitation, not a tuning failure.
RFM unpredictable from specs	Loyalty behaviour is a customer-level trait independent of the device. RFM fields in the dashboard represent population priors, not phone-level predictions.
PMF label is a proxy	The constructed PMF label approximates market fit from post-launch signals. If the business has a specific definition of success (e.g. revenue target exceeded, product survived 18 months), the formula in `pillar4_pmf.build_labels()` should be updated and the model retrained.
Similarity engine assumes catalogue overlap	For a truly novel device (new form factor, unprecedented price-spec combination), the nearest neighbours may not be genuine comparators. Retrieval quality degrades when the new phone lies far from any existing reference point.
News requires internet access	The `/api/news` endpoint depends on live DuckDuckGo search. Offline or restricted-network deployments will not return articles.